

spectral_projection_locus.m

User Guide & Data Derivation Reference

Step 2 of the IPV Benchmarking Pipeline · SEE 894 · Simon Fraser University · Spring 2026

Purpose

This script is Step 2 of the IPV benchmarking pipeline. It takes a validated device (one that passed all three checks in `ipv_validation.m`) and projects its J–V curve onto four standardised corner LED spectra. The result is a PCE vs $\lambda_{\text{onset},95\%}$ locus — a quadrilateral in performance space that encodes how the device would perform under any realistic indoor LED, independent of the specific light source used in the original paper.

1. How This Script Fits Into the Pipeline

Step	What Happens
Step 1 — <code>ipv_validation.m</code>	Validates that the three digitised curves (SPD, EQE, J–V) are internally consistent. Produces J_{sc} , V_{oc} , FF, PCE, E_{tot} , and $\lambda_{\text{onset},95\%}$ for the source LED. If APPROVED, proceed to Step 2.
Step 2 — <code>spectral_projection_locus.m</code> (this script)	Loads the same three digitised CSVs plus the four corner LED spectra. Uses the EQE to calculate spectral responsivity $R(\lambda)$. Projects the J–V curve onto each corner LED using the Khampa et al. spectral projection formula. Computes PCE at each corner and plots the locus quadrilateral.
Step 3 — <code>combined_locus_final.m</code> (separate)	Overlays all approved device loci on a single figure for cross-technology comparison.

Prerequisite

Only run this script for a paper that passed ALL THREE checks in `ipv_validation.m`. Using unvalidated data will produce a locus that is not physically meaningful.

2. File Requirements

You need five CSV files, all in the **same folder** as the script:

File	Description
<code>CIE_sle_photopic.csv</code>	Standard CIE photopic luminosity function. Shared across all papers — do not change. Same file used in Step 1.
<code>LED_spectra.csv</code>	The four corner reference LED spectra. Shared across all papers. Column layout: Col 1 = wavelength (nm), Col 2 = 5700K std (Ref 4, 644 nm), Col 3 = 3000K FS (Ref 3, 660 nm), Col 4 = 5700K FS (Ref 2, 702 nm), Col 5 = 3000K KSF (Ref 1, 721 nm).
<code>spd_file</code> (e.g. <code>s3_spd.csv</code>)	Exactly the same SPD CSV used in Step 1 for this paper.

ege_file (e.g. s3_ege.csv)	Exactly the same EQE CSV used in Step 1.
jv_file (e.g. s3_jv.csv)	Exactly the same J–V CSV used in Step 1.

3. User-Configurable Parameters

Edit only the block labelled **CHANGE THESE LINES FOR EACH PAPER**. All other sections should remain unchanged.

Parameter	How to Set It
paper_label	Short identifier used in console output and figure titles (e.g. 's3', 's12'). Must match the label you used in Step 1 for consistency.
paper_info	A second descriptive string added as a subtitle to the locus figure (e.g. 's5' or 'Perovskite 1.67 eV'). Can be anything — purely cosmetic.
f_spd, f_ege, f_jv	Filenames of the three CSVs for this paper. Must be in the same folder as the script.
SPD_UNITS	Must be set to the same value you used in Step 1 for this paper. See Section 4 for all valid values.
E_ref_uWcm2	Only required when SPD_UNITS = 'au_rescale'. Set to the total irradiance ($\mu\text{W}/\text{cm}^2$) reported in the paper.
locus_color	RGB triplet [R G B] controlling the fill and edge colour of the locus quadrilateral in Figure 4. Use the technology colour from the comment block (see below).

Technology colour guide

```
locus_color = [0.12 0.47 0.71] % Perovskite — blue
locus_color = [0.89 0.47 0.20] % OPV — orange
locus_color = [0.17 0.63 0.17] % DSSC — green
locus_color = [0.58 0.40 0.74] % a-Si:H — purple
locus_color = [0.84 0.15 0.16] % Selenium — red
```

4. SPD Unit Flags

The SPD_UNITS flag must match exactly what you used in ipv_validation.m for this paper.

SPD_UNITS value	When to use	Conversion applied
'uW/cm2/nm'	Most common. Axis labelled $\mu\text{W cm}^{-2} \text{ nm}^{-1}$.	$\div 100 \rightarrow \text{W}/\text{m}^2/\text{nm}$
'mW/m2/nm'	Axis labelled $\text{mW m}^{-2} \text{ nm}^{-1}$.	$\div 1000 \rightarrow \text{W}/\text{m}^2/\text{nm}$
'W/m2/nm'	Already in SI units.	No change
'photon_flux'	Axis labelled $\times 10^{12} \text{ cm}^{-2} \text{ s}^{-1} \text{ nm}^{-1}$.	$\times 10^{12} \times 10^4 \times h\nu/\lambda \rightarrow \text{W}/\text{m}^2/\text{nm}$
'photon_flux_e13'	Axis labelled $\times 10^{13} \text{ cm}^{-2} \text{ s}^{-1} \text{ nm}^{-1}$.	$\times 10^{13} \times 10^4 \times h\nu/\lambda \rightarrow \text{W}/\text{m}^2/\text{nm}$
'photon_flux_e15'	Axis labelled $\times 10^{15} \text{ cm}^{-2} \text{ s}^{-1} \text{ nm}^{-1}$.	$\times 10^{15} \times 10^4 \times h\nu/\lambda \rightarrow \text{W}/\text{m}^2/\text{nm}$

'au_rescale'	Arbitrary units. Requires E_ref_uWcm2.	Rescale to $\int E \, d\lambda = E_{\text{ref}}/100 \text{ W/m}^2$
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5. The Core Physics: Spectral Projection

5.1 The Spectral Responsivity $R(\lambda)$

The spectral responsivity converts irradiance at each wavelength into photocurrent density. It is derived from the EQE:

$$R(\lambda) = \text{EQE}(\lambda) \times q \times \lambda[\text{m}] / (h \times c) \quad [\text{A/W}]$$

Symbol	Meaning
EQE(λ) [—]	External quantum efficiency as a fraction (0–1). Fraction of photons at wavelength λ that produce a collected electron.
q [C]	Elementary charge = $1.602 \times 10^{-19} \text{ C}$.
$\lambda[\text{m}]$ [m]	Wavelength converted from nm to metres ($\lambda_{\text{nm}} \times 10^{-9}$).
$h \times c$ [J·m]	Planck constant $\times$ speed of light. Energy per photon at wavelength λ (in J) = $h \times c / \lambda[\text{m}]$.
$R(\lambda)$ [A/W]	Current per unit incident power at each wavelength. Peak $R(\lambda)$ occurs near the EQE peak, shifted toward longer wavelengths by the λ factor.

5.2 Equal-Irradiance Rescaling of Corner LEDs

Before projection, each corner LED spectrum is rescaled so its total irradiance matches the source LED total irradiance:

$$\begin{aligned} \text{scale} &= E_{\text{tot_source}} / E_{\text{tot_corner}} \\ E_{\text{corner_rescaled}}(\lambda) &= E_{\text{corner_raw}}(\lambda) \times \text{scale} \end{aligned}$$

This is the equal-irradiance condition, not equal-lux. The distinction matters because full-spectrum LEDs (high red content) require much more total irradiance to achieve the same lux as KSF phosphor LEDs (narrow red peak). Scaling to equal lux would introduce large spurious current shifts that are physically unrepresentative.

5.3 Spectral Projection Formula

The projected J–V curve under a target LED is derived from the source J–V using:

$$\begin{aligned} \Delta J_L &= \int R(\lambda) \times [E_{\text{target}}(\lambda) - E_{\text{source}}(\lambda)] \, d\lambda \times 100 \quad [\mu\text{A}/\text{cm}^2] \\ J_{\text{target}}(V) &= J_{\text{source}}(V) + \Delta J_L \end{aligned}$$

Symbol	Meaning
$R(\lambda)$ [A/W]	Spectral responsivity computed from EQE (Section 5.1).
$E_{\text{target}}(\lambda)$ [W/m ² /nm]	Rescaled corner LED spectrum interpolated onto the common wavelength grid.
$E_{\text{source}}(\lambda)$ [W/m ² /nm]	Source LED spectrum (from the paper) interpolated onto the common wavelength grid.
ΔJ_L [$\mu\text{A}/\text{cm}^2$]	Photocurrent shift: the difference in photogenerated current between the target and source LEDs. Positive = target LED generates more current. The $\times 100$ converts A/m ² to $\mu\text{A}/\text{cm}^2$.
$J_{\text{target}}(V)$ [$\mu\text{A}/\text{cm}^2$]	The full J–V curve under the target LED. Only the photocurrent (J_{sc}) changes; V_{oc} and FF are assumed constant (first-order approximation).

Sign convention

The original Khampa et al. formula uses $J < 0$ convention. This code uses $J > 0$ (positive in the power quadrant). The shift direction is $(E_{\text{target}} - E_{\text{source}})$, which means a target with more red content than the source gives $\Delta J_L > 0$ and a higher projected current — physically correct.

5.4 PCE at Each Corner

Once the projected J–V curve is available, PCE is calculated at each corner using:

$$\begin{aligned} P_{\text{target}}(V) &= V \times J_{\text{target}}(V) && [\mu\text{W}/\text{cm}^2] \\ \text{PCE}_{\text{target}} &= \max(P_{\text{target}}) / E_{\text{tot_target}} \times 100 && [\%] \end{aligned}$$

where $E_{\text{tot_target}}$ is the total irradiance of the rescaled corner LED (which equals $E_{\text{tot_source}}$ by construction). Each of the four corners produces one $(\lambda_{\text{onset},95\%}, \text{PCE})$ coordinate.

5.5 $\lambda_{\text{onset},95\%}$ of the Corner LEDs

After rescaling, $\lambda_{\text{onset},95\%}$ is computed for each corner spectrum using the same cumulative integral method as in Step 1:

$$\int [\lambda_{\text{min}} \rightarrow \lambda_{\text{onset},95\%}] E(\lambda) d\lambda = 0.95 \times \int [\lambda_{\text{min}} \rightarrow \infty] E(\lambda) d\lambda$$

Because rescaling only changes amplitude (not spectral shape), $\lambda_{\text{onset},95\%}$ is the same for the raw and rescaled corner LEDs. The four corner values define the x-axis span of the locus:

- **Ref 4** — 5700K std LED: $\lambda_{\text{onset},95\%} \approx 644.8$ nm (shortest, most blue-rich)
- **Ref 3** — 3000K FS LED: $\lambda_{\text{onset},95\%} \approx 660.1$ nm
- **Ref 2** — 5700K FS LED: $\lambda_{\text{onset},95\%} \approx 702.0$ nm

- **Ref 1** — 3000K KSF LED: $\lambda_{\text{onset},95\%} \approx 721.1 \text{ nm}$ (longest, most red-rich)

6. Console Output Explained

The script prints two blocks. Here is an annotated example for Device A (s3):

```
=====
SPECTRAL PROJECTION - s3
=====
Source SPD (as measured):
  lambda_onset,95% = 677.2 nm      <- spectral descriptor of the paper's LED
  E_tot            = 278.38 uW/cm2 <- total irradiance of paper's LED

Jsc from EQE x SPD : 122.690 uA/cm2 <- agrees with Step 1 Check 3
Jsc from J-V       : 130.847 uA/cm2
Delta_Jsc          : 6.23%         <- same as Step 1 mismatch

Ref    LED                195 (nm)   E_tot   delta_JL   Jsc_proj   PCE
-----
Ref4   5700K (644nm)      644.8    278.38   -3.1850   127.662    40.58%
Ref3   3000K (660nm)      660.1    278.38   -2.8421   128.005    43.16%
Ref2   5700K (702nm)      702.0    278.38   +1.4260   132.273    41.18%
Ref1   3000K (720nm)      720.1    278.38   +4.5600   135.407    43.76%

=====
LOCUS COORDINATES - s3
=====
Ref4   5700K (644nm)      (644.8 nm, 40.58%)
Ref3   3000K (660nm)      (660.1 nm, 43.16%)
Ref2   5700K (702nm)      (702.0 nm, 41.18%)
Ref1   3000K (720nm)      (720.1 nm, 43.76%)

Source (meas.)            (677.2 nm, 42.81%)

PCE range : 40.58% - 43.76% (spread = 3.18%)
lambda95   : 644.8 - 720.1 nm
=====
```

Column	Meaning
195 (nm)	$\lambda_{\text{onset},95\%}$ of the rescaled corner LED. Fixed per corner — does not change between papers.
E_tot	Total irradiance of the corner LED after rescaling ($\mu\text{W}/\text{cm}^2$). Equals $E_{\text{tot_source}}$ by construction.
delta_JL	Photocurrent shift ΔJ_L ($\mu\text{A}/\text{cm}^2$). Negative = corner has less red content than source → lower Jsc. Positive = corner has more red content → higher Jsc.
Jsc_proj	Projected Jsc at each corner ($\mu\text{A}/\text{cm}^2$). Equals $J_{\text{sc_source}} + \Delta J_L$ evaluated at $V = 0$.
PCE	Projected power conversion efficiency at each corner (%). This is the y-axis of the locus.

PCE range / spread	Minimum and maximum PCE across the four corners, and their difference. A large spread indicates the device is sensitive to LED spectral variation.
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7. The Four Figures

Figure	What It Shows and What to Check
Figure 1 — Corner LED Spectra	All four corner LEDs (coloured lines) and the source SPD (black dashed), all rescaled to equal total irradiance. Check that the source SPD falls within the range of the four corners. If the source is far outside, the device was measured under an unusual LED, and the measured PCE may not be representative.
Figure 2 — EQE and R(λ)	Left axis (green): EQE (%). Right axis (blue): spectral responsivity R(λ) in A/W. R(λ) should rise with wavelength in the absorbing region because longer-wavelength photons carry less energy per photon, so the same EQE produces more electrons per watt. A flat or decreasing R(λ) in the visible suggests a data issue.
Figure 3 — Projected J–V Curves	The source J–V (black dashed) and the four projected J–V curves (coloured). The curves should be nearly parallel — the projection shifts current uniformly by ΔJ_L . A large vertical separation between curves confirms strong spectral sensitivity. If curves cross or show large non-uniformity, check EQE and SPD quality.
Figure 4 — PCE– $\lambda_{\text{onset,95\%}}$ Locus	The main deliverable. Four coloured corner points (circle, triangle, square, diamond) connected into a filled quadrilateral. The measured source point is marked with a black asterisk. The x-axis spans 630–735 nm; the four vertical dashed lines mark the four corner $\lambda_{\text{onset,95\%}}$ values. 3000K and 5700K labels are shown on the locus body.

Reading the locus shape

A wide, flat locus (large x-span, small y-variation) means the device is insensitive to LED spectral distribution — consistent performance across all realistic LEDs. A tall, narrow locus means the device is strongly affected by spectral shape. A tilted or asymmetric locus (e.g. Device B at 1.79 eV) indicates the bandgap is mismatched for some corner conditions.

8. Derived Data: What Each Output Means

Quantity	Physical Meaning	How to Use It
R(λ) [A/W]	Spectral responsivity: current generated per watt of incident light at each wavelength. Derived from EQE.	Intermediate quantity only — used internally for the projection integral. Plotted in Figure 2.
ΔJ_L [$\mu\text{A}/\text{cm}^2$]	Photocurrent shift from source to corner LED. Captures the spectral mismatch.	Positive = target richer in long-wavelength photons. Record for reporting if needed. Printed per corner in the table.

J_target(V) [$\mu\text{A}/\text{cm}^2$]	Complete J–V curve projected onto the target corner LED. Only Jsc shifts; Voc and FF are unchanged.	Plotted in Figure 3. Used to compute PCE at the corner.
PCE_corner [%]	Power conversion efficiency under each of the four corner LEDs.	The y-axis of the locus. Record all four for the combined locus script.
$\lambda_{\text{onset},95\%_{\text{corner}}}$ [nm]	Spectral descriptor of each corner LED.	The x-axis of the locus. Fixed values: 644.8, 660.1, 702.0, 721.1 nm. Do not recalculate per paper.
PCE_source [%]	Recalculated PCE under the paper's source LED. Should agree closely with PCE_calc from Step 1.	Cross-check: if PCE_source from this script differs significantly from Step 1, check that SPD_UNITS and filenames match exactly.
PCE spread [%]	$\max(\text{PCE_corners}) - \min(\text{PCE_corners})$.	Measures sensitivity to LED spectral variation. Large spread = bandgap is near the optimal for some conditions but not others.
Locus quadrilateral	The region of PCE– λ space achievable by any LED within the realistic indoor range.	Plotted as a filled quadrilateral in Figure 4. Recorded for Step 3 (combined locus).

9. Step-by-Step Usage

1. **Confirm Step 1 passed.** The paper must have PASSED all three checks in ipv_validation.m. Do not proceed otherwise.
2. **Copy the same three CSV files.** Use the identical SPD, EQE, and J–V files that you used in Step 1. Do not re-digitise.
3. **Confirm LED_spectra.csv is present** in the same folder. This file is shared across all papers and should already be there.
4. **Open spectral_projection_locus.m** and edit only the top configuration block.
5. **Set paper_label** to the same identifier used in Step 1 (e.g. 's3').
6. **Set paper_info** to any descriptive subtitle (e.g. the absorber class and bandgap).
7. **Set f_spd, f_eqe, f_jv** to your CSV filenames.
8. **Set SPD_UNITS** to exactly the same value used in Step 1 for this paper.
9. **Set locus_color** to the RGB triplet for the technology class (see colour guide in Section 3).
10. **Run the script.** Read the console output. Verify that the Jsc from EQE $\times$ SPD and Jsc from J–V agree ($\Delta_{\text{Jsc}} \leq 10\%$). These should match Step 1 Check 3.
11. **Inspect the four figures.** Check Figure 3 (projected J–V curves look parallel) and Figure 4 (locus shape is reasonable).
12. **Record the locus coordinates:** four ($\lambda_{\text{onset},95\%}$, PCE) pairs plus the measured source point. These are passed to the combined locus script.

10. Corner LED Column Reference

The LED_spectra.csv file must have exactly this column layout:

Column	Reference Label	Description
Col 1	—	Wavelength in nm (shared x-axis for all four LEDs)
Col 2	Ref 4 5700K std	5700K standard phosphor LED. $\lambda_{\text{onset},95\%} \approx 644.8$ nm. Shortest spectral reach — most blue-dominant.
Col 3	Ref 3 3000K FS	3000K full-spectrum LED. $\lambda_{\text{onset},95\%} \approx 660.1$ nm.
Col 4	Ref 2 5700K FS	5700K full-spectrum LED. $\lambda_{\text{onset},95\%} \approx 702.0$ nm.
Col 5	Ref 1 3000K KSF	3000K KSF phosphor LED. $\lambda_{\text{onset},95\%} \approx 721.1$ nm. Longest spectral reach — most red-rich.

Corner ordering note

In the locus plot, the quadrilateral is drawn in the order Ref4 → Ref2 → Ref1 → Ref3 → close (i.e. quad_idx = [1,3,4,2]). This connects the 5700K pair and the 3000K pair as the two diagonals, producing the correct parallelogram shape. The Ref numbering is from the project report, not the column order.

11. Troubleshooting

Symptom	Likely Cause and Fix
Locus looks like a single point (all four PCEs nearly identical)	Check that LED_spectra.csv has four distinct LED spectra. Also check that $R(\lambda)$ is non-zero — if the EQE is zero everywhere, $\Delta J_L = 0$ for all corners.
PCE_source here differs significantly from Step 1 PCE_calc	SPD_UNITS is set differently between scripts, or you are using a different CSV file. Ensure both scripts use identical settings.
Projected J–V curves cross each other	A crossing indicates the spectral shift is not uniform across voltage — this can happen if the J–V curve has digitisation noise. Smooth the J–V data and re-run.
ΔJ_L is extremely large ($> 50 \mu\text{A}/\text{cm}^2$)	The source LED is far outside the realistic corner range ($\lambda_{\text{onset},95\%}$ much less than 644 nm or much greater than 721 nm). The projection is technically valid but the locus will be large and the device's performance may be unrepresentative. Flag this in the report.
Figure 4 y-axis range is very narrow (all corners nearly same PCE)	Small locus spread is physically real for some devices (e.g. OPV with EQE spread across visible). Verify by checking Figure 3 — if J–V curves are nearly overlapping, the result is correct.
Error: LED_spectra.csv not found	Place LED_spectra.csv in the same folder as the script and all CSV files.
Error: Index exceeds matrix dimensions	Usually caused by a J–V CSV with fewer than two columns, or by V_s and J_s having different lengths after clipping. Check the J–V CSV for blank rows or headers.

12. Physical Constants Used

Constant	Value and Role
h_p (Planck constant)	$6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ — used in $R(\lambda)$ and photon energy calculations
c_l (Speed of light)	$3.000 \times 10^8 \text{ m/s}$ — used in $R(\lambda)$ and photon energy calculations
q (Elementary charge)	$1.602 \times 10^{-19} \text{ C}$ — converts photon flux to current density in $R(\lambda)$
$\times 100$ conversion factor	$1 \text{ A/m}^2 = 100 \text{ }\mu\text{A/cm}^2$. Applied to ΔJ_L integral to convert from SI to $\mu\text{A/cm}^2$.

13. Data to Record After Running

Copy the following from the LOCUS COORDINATES summary block into your data record for Step 3 (combined_locus_final.m):

Value to Record	Where to Find It
PCE at Ref 4 (644.8 nm)	Console: Ref4 row, PCE column
PCE at Ref 3 (660.1 nm)	Console: Ref3 row, PCE column
PCE at Ref 2 (702.0 nm)	Console: Ref2 row, PCE column
PCE at Ref 1 (721.1 nm)	Console: Ref1 row, PCE column
PCE at source (measured)	Console: Source (meas.) line
$\lambda_{\text{onset},95\%}$ at source	Console: Source SPD block, $\lambda_{\text{onset},95\%}$
PCE spread (%)	Console: PCE range line
locus_color used	Your configuration block

14. Reference

The spectral projection method (Note S8) is from:

Khampa et al. (2026). *"A locus-based benchmarking framework for indoor photovoltaics."*
Newton (in press).

Project report: *"A Universal Design and Benchmarking Framework for Indoor Photovoltaics."*
Olarewaju Okikioluwa Victor & Ali Ghahremani. SEE 894, Simon Fraser University, Spring 2026.

This guide was generated from the source code of spectral_projection_locus.m. All formulas, sign conventions, and parameter descriptions reflect the actual implementation.